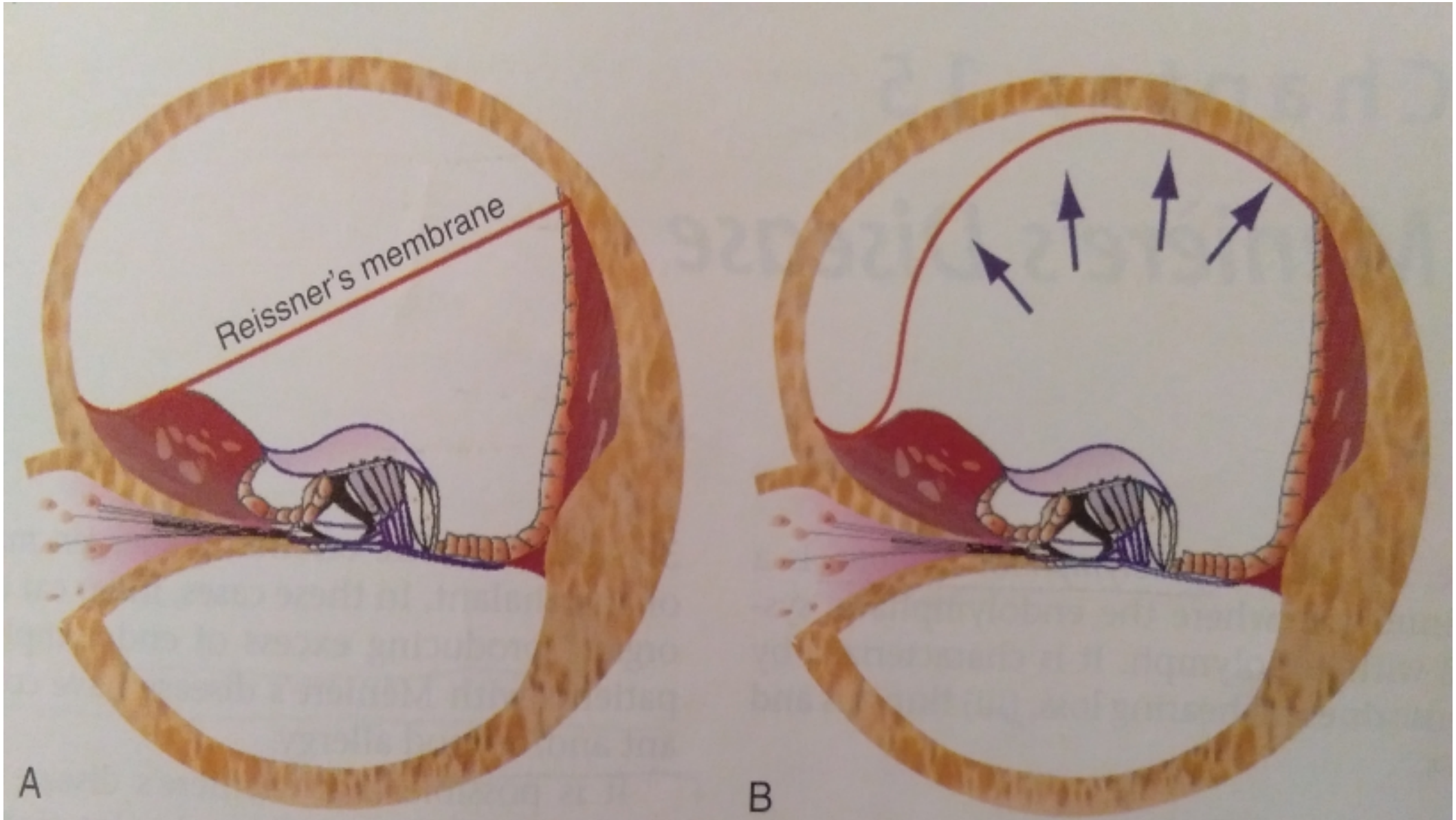
# Meniere's Disease

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# Pathology

- Endolymphatic hydrops
- Distension of Endolymphatic system esp. Cochlear duct and saccule.
- Bulging of Reissner 's membrane
- Dilation of saccule




ormal cochlear duct. (B) Cochlear duct is distended with endolymph pushing the Reissner's membrane ir

# Etiology and Symptamatology

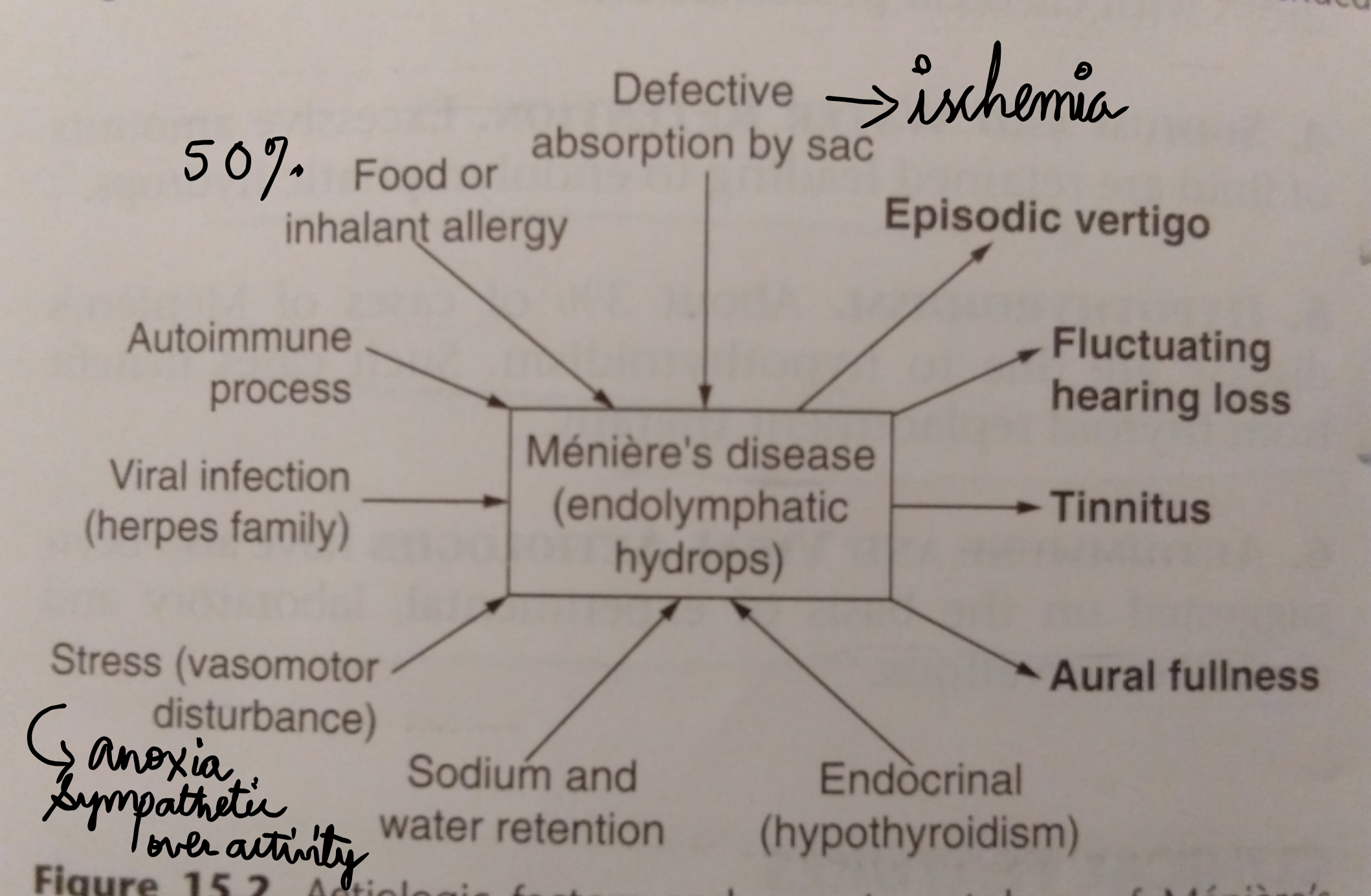
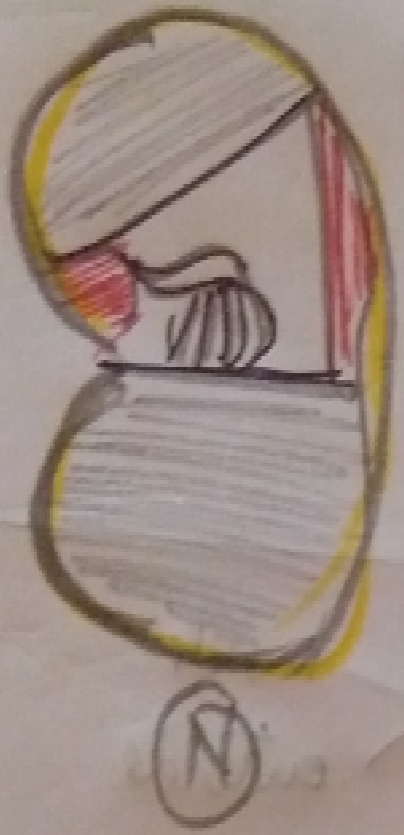



**Figure 15.2.** Aetiologic factors and symptomatology of Ménière's disease (endolymphatic hydrops).

# Symptoms

- **Vertigo**

1. Transient, sudden, episodic (20 min-24 hrs)
2. Associated with vomiting, nystagmus
3. Tullios phenomenon
4. Drop attacks/ Tumarkin crisis






Menière's

↑ endolymph → SM dilated

↓

Reissner's Membrane bulges

↓

Repeated prodn of endolymph → vol ↑

↓

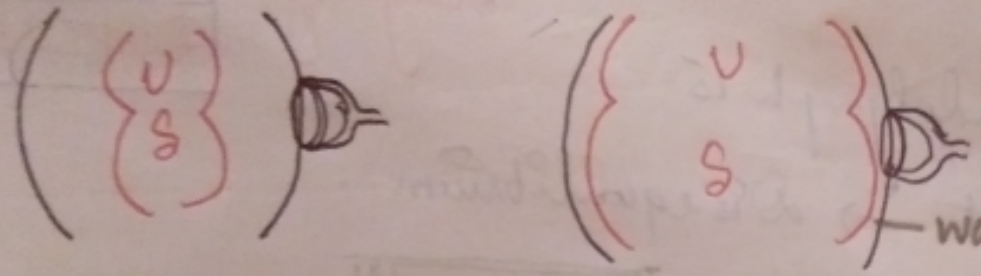
Reissner's M. keeps bulging → Finally RUPTURES

⇒ Sudden gushing of endolymph to Perilymph compartment ⇒ Disequilibrium.

Consequence

1st complaint of Menière's ⇒ Vertigo

## Changes in Saccule and Utricle → Dilated < inside / outside




Normal Dilated

Dilation inside
→ Damage to maculae
∴ Drop crisis / Tumarkin crisis

Dilated outside
→ In case of very loud sounds footplate of stapes will move more inwards and touches U / Saccule causes vertigo.
∴ Tullios phenomenon.

- **Hearing loss** -

1. fluctuating sensorineural type
2. Binaural diplacusis
3. Recruitment phenomenon

- Tinnitus - subjective, warning sign
- Aural fullness - fluctuating
- Others- anxiety, stress

# Other variants

- Cochlear hydrops - no vertigo,  block at Ductus reuniens
- Vestibular hydrops -no hearing loss
- Lermoyez syndrome - rare,  symptoms occur in reverse order

# Investigations

- Pure tone audiometry

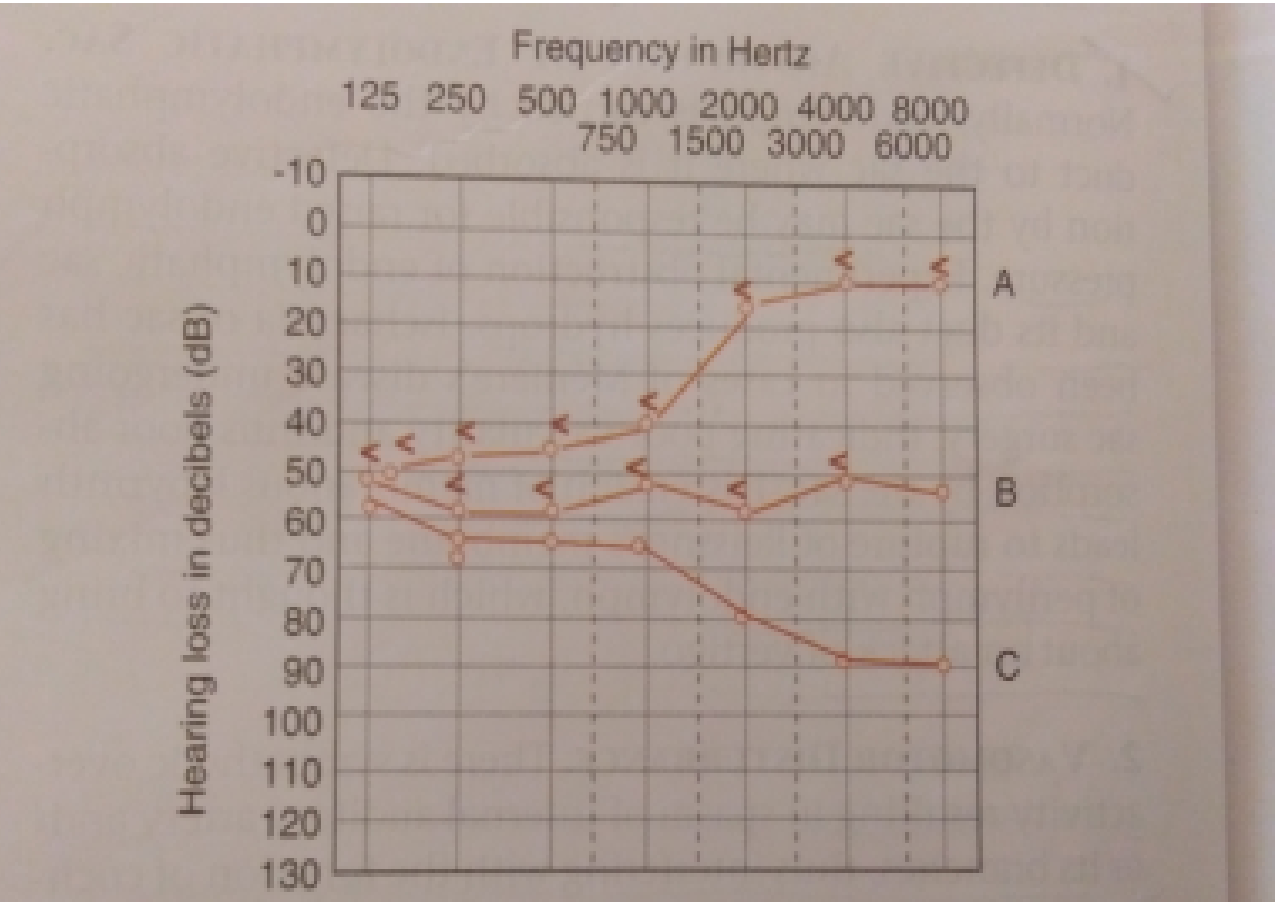



**Figure 15.3.** (A) Audiogram in early Ménière's disease. Note: Hearing loss is sensorineural and more in lower frequencies—the rising curve. As the disease progresses, middle and higher frequencies get involved and audiogram becomes flat or falling type (B & C).

- Sensorineural hearing loss
- Early Meniere's - low frequency affected - rising type
- Later stages- higher frequency affected - falling type

# Special tests

- Recruitment test +
- SISI score >70%
- Tone decay test less than 25 dB ( N<20 dB)

# Electrocochleography

- SP:AP >30%
- ( avg 45% )
- [Normal =30%]

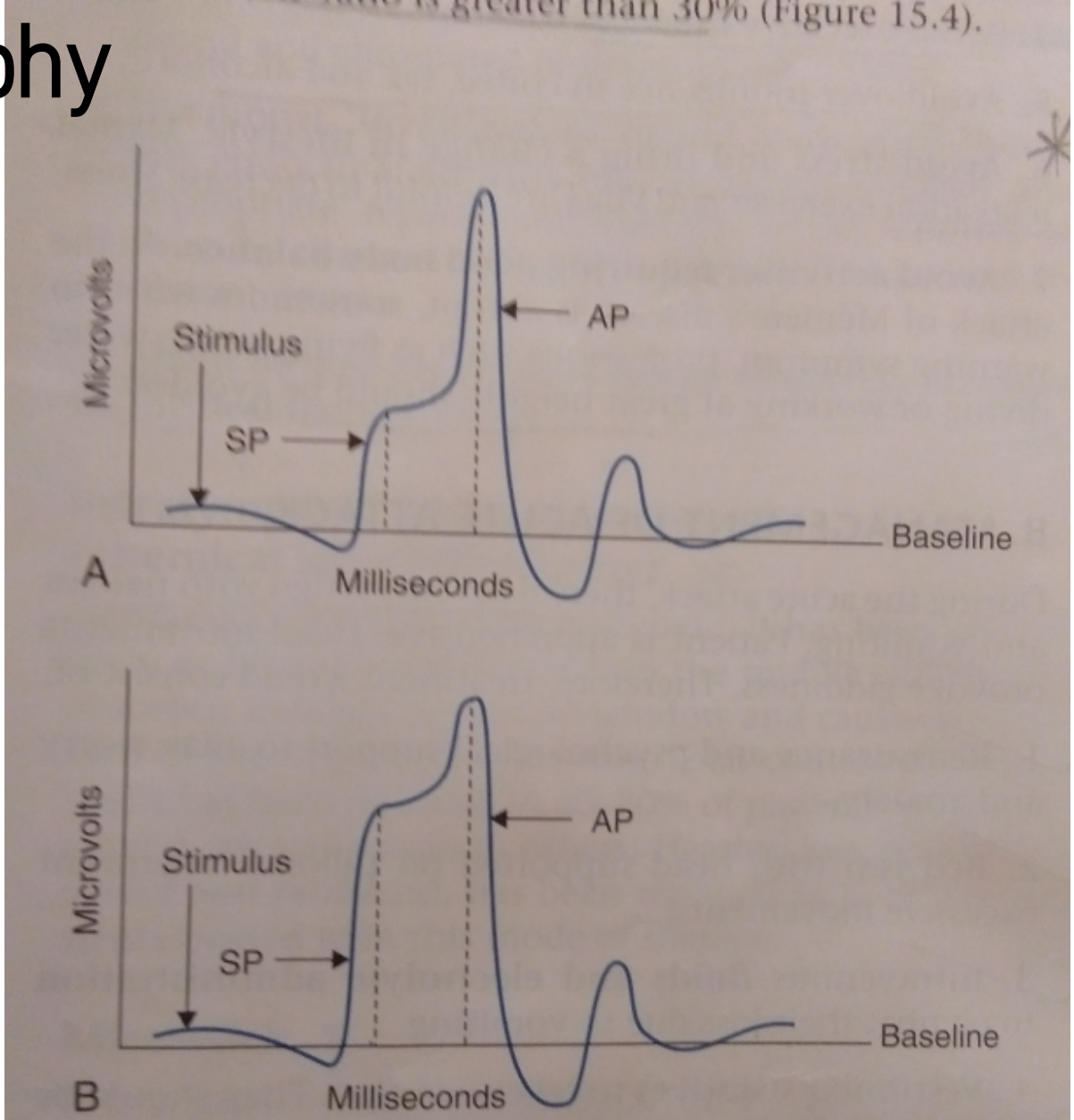



**Figure 15.4.** Electrocochleography. (A) Normal ear. (B) Ear with Ménière's disease. Voltage of summating potential (SP) is compared with that of action potential (AP). Normally SP is 30% of AP. This ratio is enhanced in Ménière's disease.

- Caloric test reduced on affected ear in 75% cases

- Glycerol test - dehydrating agent

- 1.5 ml/kg with equal amount of water + flavouring agent given orally
- Audiogram recorded 1-2 hrs later
- Improvement of 10 dBs in 2 or more adjacent octaves is +ve test
- Combined with electrocochleography

# Treatment

# General

- Reassurance of the patient
- Cessation of smoking
- Low salt diet
- Avoid excess water intake
- Avoid caffeinated drinks
- Avoid stress
- Lifestyle modifications
- Avoid activities requiring good body balance - flying, diving, etc.

# Acute attacks

- Reassurance
- Bed rest
- IV fluids and electrolytes to replenish sources after vomiting
- Vestibular sedatives - dimenhydrinate, promethazine, prochlorperazine.
- IV Diazepam 5-10 MG Tranquilizing effect
- Vasodilators- eg. Carbogen

# Chronic phase

- Vestibular sedatives - prochlorperazine
- Vasodilators - Tab. Betahistine 8 - 16 mg TID orally
- Diuretics like Frusemide
- Avoiding allergens
- Hormones - in case of hypothyroidism

# In case of intractable vertigo

- Intratympanic gentamicin therapy / chemical labyrinthectomy - vestibulotoxic -through round window - microwicks

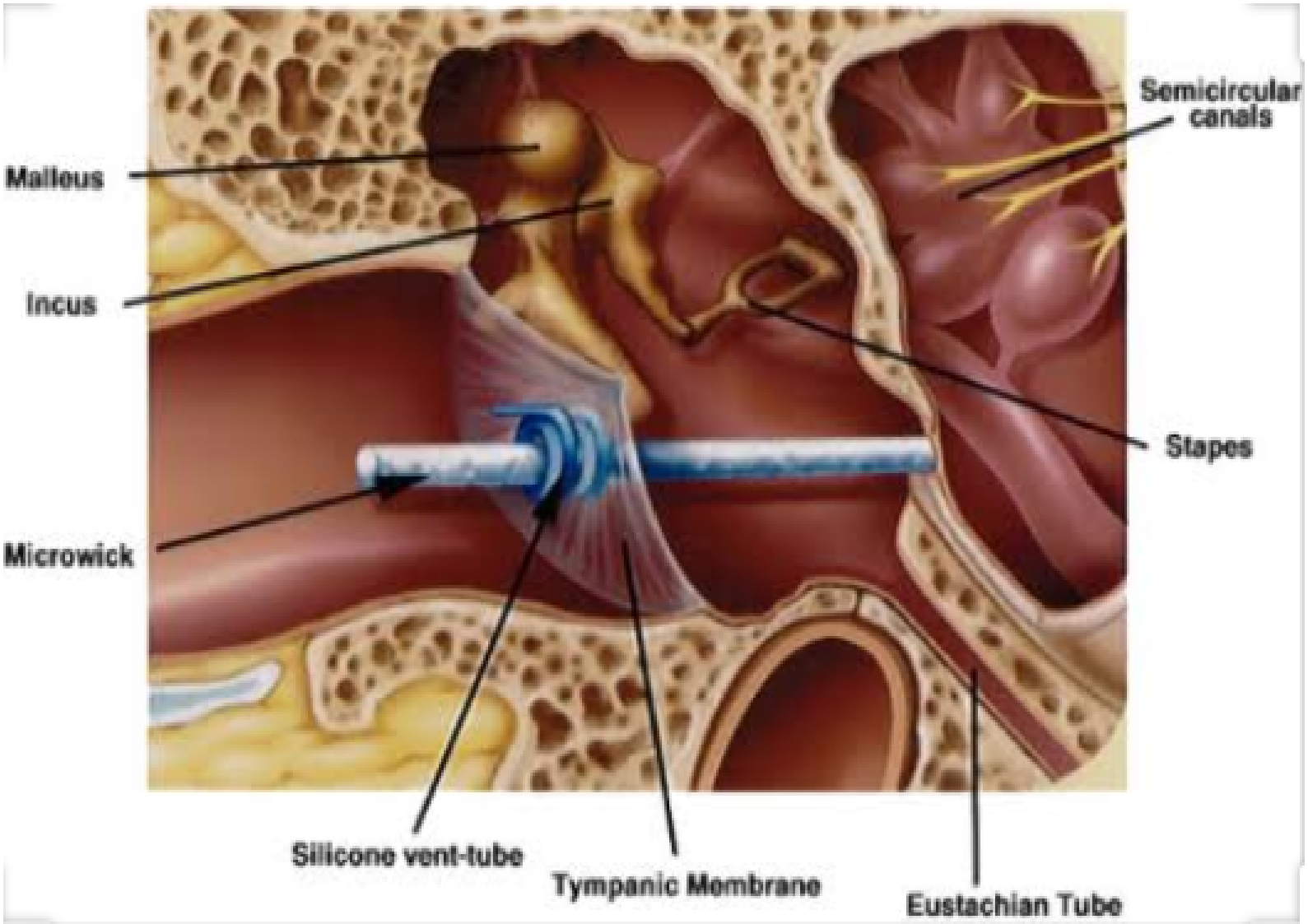



Image source - blog.earsinus.com

# Surgical Treatment

■Conservative management

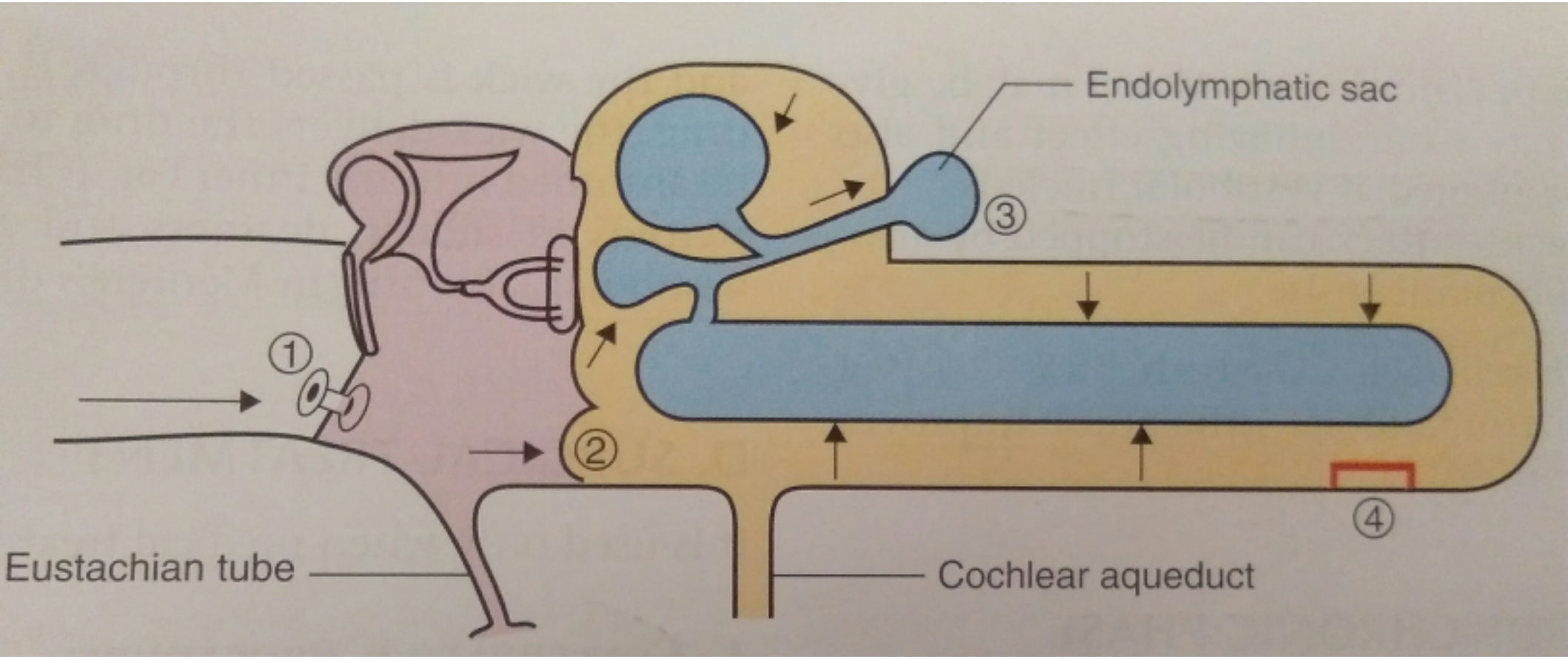
1. Decompression of Endolymphatic sac
2. Endolymphatic shunt operation to subarachnoid space
3. Fick's operation /sacculotomy
4. Selective Vestibular Neurectomy
5. Ultrasonic destruction of vestibular labyrinth

■ Destructive procedure

1. Labyrinthectomy

■ Meniett device therapy

Delivery of positive pressure to round window through this device.



# THANK YOU
# Any queries - please contact 8197741790

Reference -

PL Dhingra 7th edition